

Enterprise Infrastructure Plan

ABC Startup Solutions

Designing an enterprise IT infrastructure from scratch
for a 20-employee software development startup

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Academic disclaimer. ABC Startup Solutions is a fictional company created for this academic project. The company profile, office address, staffing and asset records in this document are illustrative. No equipment has been purchased, no service provider has been engaged, and no part of this design has been deployed.

Executive Summary

ABC Startup Solutions is a newly established software development company with 20 employees occupying a single leased office floor. At the time of this plan the company has no computers, no server, no network, no internet service and no security policies. This document sets out a complete enterprise infrastructure plan that takes the company from an empty floor to a working, secured and documented IT environment.

The proposed design is deliberately proportionate to a 20-person business. A single virtualisation server, one managed switch, one edge router, one UTM firewall and two wireless access points carry the entire office. Email, identity, file collaboration and endpoint management are placed in Microsoft 365 Business Premium rather than on-premises Windows servers, which removes an entire class of licensing, patching and backup work from a company that has exactly one IT Manager. Everything that must remain on site — source code, development and staging systems, internal tooling and backup data — runs on one Ubuntu Server host and one NAS.

The network is segmented into six VLANs so that Finance and HR data is separated from general office traffic, with all inter-department routing forced through the firewall where it can be filtered and logged. Guest wireless is isolated to internet access only. Data protection follows a 3-2-1 backup strategy combining NAS snapshots, weekly external drives rotated off site, and cloud copies, with a stated 24-hour recovery point objective and quarterly restore testing.

Total planned equipment comes to 55 units across 12 asset classes, with 21 of the 48 switch ports and roughly half the server's memory left free. That headroom is intentional: the same core equipment supports growth to approximately 35 staff before any component must be replaced, which is set out in the expansion plan in Part 7.

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Part 1 — Company Profile

Business identity, organisational structure and the infrastructure requirements that follow from them

1.1 Company Name and Nature of Business

Company name. ABC Startup Solutions

ABC Startup Solutions is a newly established software development company that designs, builds and maintains custom web and mobile applications for small and medium enterprises in the Philippines. Alongside project delivery, the company provides post-deployment application support and light IT consulting to its clients. Because it is a services business, its most valuable assets are its people, its source code and the client data entrusted to it — all three depend directly on the reliability and security of the internal infrastructure planned in this document.

1.2 Vision

To become a trusted software development partner for growing Philippine businesses, recognised for building dependable systems on a secure and well-managed technology foundation.

1.3 Mission

- Deliver custom software that solves real operational problems for small and medium enterprises.
- Maintain a secure, reliable and well-documented IT environment that protects both client and company data.
- Invest in the technical growth of our people through continuous learning and structured mentoring.
- Grow sustainably by choosing technology that fits the company we are today and scales with the company we intend to become.

1.4 Office Location

Address	Unit 3A, Meridian Corporate Centre, 128 Pioneer Business Park, Barangay San Antonio, Santa Cruz, Laguna 4009, Philippines
Premises	One leased office floor of approximately 240 square metres
Layout	Open-plan work area, one meeting room, one reception area, and one lockable utility room used as the server and communications cabinet location
Occupancy	20 employees, single shift, standard office hours

Table 1 — Premises summary.

1.5 Organisational Structure

The company runs a flat, two-level structure appropriate to its size. A Managing Director provides overall direction, and four department heads report directly to that role. The Managing Director is a founding partner who concurrently heads Sales & Business Development and is counted within the six-person Sales headcount, so total staffing remains exactly 20.

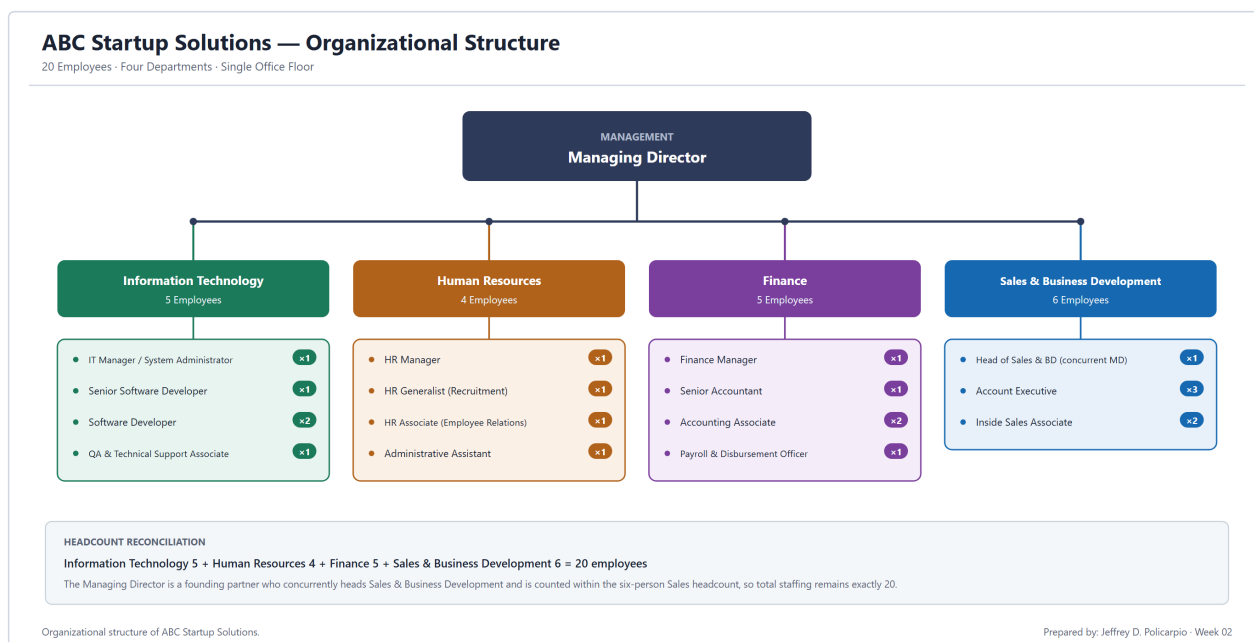


Figure 1 — Organisational structure of ABC Startup Solutions.

1.6 Employee Distribution

Department	Employees	Share	Primary Device Allocation
Information Technology	5	25%	4 desktop workstations (developers and QA) plus 1 laptop for the IT Manager
Human Resources	4	20%	4 desktop workstations
Finance	5	25%	5 desktop workstations, each on a dedicated UPS
Sales & Business Development	6	30%	4 laptops for client-facing staff plus 2 desktop workstations for inside sales
Total	20	100%	15 desktops, 5 assigned laptops, 1 spare laptop

Table 2 — Employee distribution and primary device allocation.

1.7 How the Company Structure Shapes IT Requirements

Information Technology (5) drives the compute requirement.

The developers and QA associate run local builds, containers and virtual machines. They need more memory, faster storage and elevated local privileges than any other group, and they are the only staff who require VirtualBox and access to the server's management interfaces. Their needs set the specification for the highest-tier workstation.

Finance (5) and Human Resources (4) hold the most sensitive data.

Payroll records, employment contracts, government filings and client billing information sit with these two departments. Under the Data Privacy Act of 2012 (Republic Act 10173) the company is accountable for protecting personal information it processes. This is the single strongest argument for segmenting the network by department rather than running one flat LAN, and for placing Finance workstations on dedicated power protection.

Sales & Business Development (6) is the most mobile group.

Four of the six work off site with clients, which makes laptops, reliable wireless coverage and secure remote access into the office network a functional requirement rather than a convenience.

A 20-person headcount means one generalist, not a team of specialists.

The company can justify exactly one IT Manager / System Administrator. That person cannot specialise, so the design must favour manageability over sophistication: few devices, central management, cloud-hosted identity and email, and configuration that is written down. Every component in this plan was checked against the question of whether one person can realistically operate it.

A flat structure on a single floor still needs logical separation.

Physically the office is one open space, and the simplest possible network would be a single flat subnet. The confidentiality requirements above justify the additional effort of VLAN segmentation, which costs nothing in hardware because the chosen switch is already managed.

Part 2 — Enterprise Hardware Inventory

Planned hardware assets, quantities and the reasoning behind them

Asset ID	Hardware	Qty	Department / Location	Purpose
HW-PC-001	Desktop Computer (business tower)	15	IT 4 · HR 4 · Finance 5 · Sales 2	Primary workstation for fixed-desk roles. Developer units carry a higher CPU, 32 GB RAM and NVMe storage; administrative units use a standard 16 GB configuration.
HW-LAP-001	Business Laptop (14")	6	IT 1 · Sales 4 · Spare 1	Mobility for the IT Manager and client-facing sales staff. One unit is held unassigned as a loaner so a hardware failure does not idle an employee during warranty repair.
HW-MON-001	24" IPS Monitor (1920×1080)	19	All departments	One screen per desktop, a second screen for each of the three developers, and one external screen for the IT Manager's docked laptop.
HW-SRV-001	Tower/Rack Server (virtualisation host)	1	IT · Server cabinet	Single Ubuntu Server host running the internal virtual machines: source control, build agent, development and staging environments, monitoring and the internal service desk.
HW-NAS-001	4-Bay NAS, 4 × 4 TB (RAID 5, ≈12 TB usable)	1	IT · Server cabinet	Departmental file shares and the on-site backup target for the server's virtual machines. Provides fast local restore without touching off-site media.
HW-BKP-001	8 TB External USB Backup Drive	2	IT · One on site, one off site	Weekly full backup written to an external drive; the two drives are rotated so one copy is always physically disconnected and stored away from the office.
HW-PRN-001	A4 Colour Laser Multifunction Printer	1	Shared · Open work area	Shared network printing, scanning and copying with PIN-based secure print release so payroll and contract documents are only produced when the owner is at the device.
HW-UPS-001	3 kVA Rack-Mount Line-Interactive UPS	1	IT · Server cabinet	Protects the server, NAS, switch, firewall, router and modem, and provides runtime for an automated graceful shutdown during an extended outage.
HW-UPS-002	1 kVA Desktop UPS	5	Finance	Protects accounting workstations, where an abrupt power loss during a posting or payroll run risks corrupting open financial records.
NET-RTR-001	Business Edge Router (dual WAN)	1	IT · Server cabinet	Terminates the fibre service, performs NAT and hosts the site VPN. A second WAN port carries the 5G/LTE failover connection. Detailed in Part 4.
NET-SW-001	48-Port Managed Gigabit PoE+ Switch	1	IT · Server cabinet	Single wiring-closet switch providing all desk drops, 802.1Q VLAN tagging and PoE+ power for the access points. Detailed in Part 4.
NET-AP-001	Wireless Access Point, Wi-Fi 6 (802.11ax), PoE+	2	Open work area · Meeting room	Wireless coverage for laptops, mobile devices and the isolated guest network. Detailed in Part 4.
	Total units	55		

Table 3 — Enterprise hardware inventory (55 units across 12 asset classes).

2.1 Asset Identifier Convention

Asset identifiers follow a fixed three-part pattern: a category prefix, a device-type code and a three-digit sequence. HW- covers end-user and server hardware, NET- covers network infrastructure. Network devices

keep their NET- identifier in both Part 2 and Part 4 rather than being given a second number, so a single device is never recorded twice under two different identities. The scheme leaves room for 999 units of each type, which the company will not exhaust.

2.2 Hardware Selection Rationale

Endpoint allocation: 21 units for 20 staff, not 20 for 20

Fifteen employees work at a fixed desk and receive desktop towers, which cost less per unit of performance, are easier to open and repair, and are typically kept in service longer than laptops. Laptops are issued only where mobility is genuinely required: the IT Manager, who works at the cabinet and in meeting rooms as often as at a desk, and the four client-facing sales staff. The twenty-first unit is an unassigned spare laptop. At roughly a five per cent spare ratio it is inexpensive, and it means a failed machine costs an employee an hour rather than the several days a warranty repair actually takes.

Monitors: 19 rather than a flat 20

Fifteen screens serve the desktop fleet, three second screens go to the developers, where working across an editor, a terminal and a browser otherwise forces constant window switching, and one screen serves the IT Manager's docked laptop. The spare laptop has its own display, so no spare monitor is needed. Rounding this figure up to 20 would have bought a monitor with no desk to sit on.

Server: one host, and the reasoning for not buying two

A second server would roughly double the hardware cost and the administrative effort for a company whose genuinely business-critical systems — email, identity, documents and collaboration — already sit in Microsoft 365 and are not affected by an on-premises outage. What remains on site is development and internal tooling, where a day of downtime is disruptive but not existential. Resilience is therefore purchased through disciplined backup and a documented rebuild procedure instead of redundant hardware. A second host is the first item in the expansion plan in Part 7.7, at the point where internal systems start carrying production client workloads.

Storage and backup: NAS plus rotating external drives

The NAS gives fast on-site recovery for the common cases, which are a deleted file or a failed virtual machine. The two external drives cover the case the NAS cannot: a fire, a theft, or ransomware that reaches every device the server can write to over the network. Because the off-site drive is physically unplugged, it cannot be encrypted by an attacker who has compromised the network.

Power protection: sized to the actual risk, not to every desk

The rack UPS protects the single point of failure in the building. Five desktop units go to Finance, where unsaved work carries the highest cost. The remaining workstations are deliberately left unprotected because their work is saved to OneDrive continuously and a brownout costs a few minutes rather than a corrupted ledger. The laser printer is deliberately excluded from any UPS: its fuser draws a large inrush current at warm-up that can overload the unit and shorten its battery life.

Wireless: two access points on one floor

A single access point can technically cover 240 square metres. A second removes the dead spot behind the meeting room and utility wall, halves the number of clients competing for airtime on one radio, and leaves partial coverage standing if one unit fails. Both are powered over PoE+ from the switch, so no electrical work is required at ceiling level.

Printing: one device with secure release, not one per department

Twenty people on a single floor are comfortably served by one multifunction device. The obvious objection is confidentiality — payroll and contracts should not sit in an open output tray — and that is solved with PIN-based secure print release rather than by buying and maintaining a second printer with its own consumables and driver estate.

Part 3 — Enterprise Software Inventory

Standard software build, licensing model and the reason each item is present

Software	Version	Licence	Purpose and Justification
Windows 11 Pro	Current supported release	OEM, per device	Standard desktop and laptop operating system. The Pro edition is required rather than Home because the company depends on BitLocker drive encryption, Group Policy and Microsoft Entra ID join, none of which Home provides.
Ubuntu Server LTS	24.04 LTS (Noble Numbat)	Free and open source	Operating system for the single virtualisation host, running KVM/libvirt. Chosen over the newer 26.04 LTS because 24.04 has two years of production history and standard security maintenance to 2029; 26.04 LTS is the planned next upgrade once it has matured.
Microsoft 365 Business Premium (includes Microsoft Office)	Current supported release	Subscription, 20 seats	Supplies the Microsoft Office applications, Exchange email, OneDrive and SharePoint, and Teams. It also bundles Microsoft Entra ID P1, Microsoft Intune and Defender for Business, which is why it was chosen over the cheaper Business Standard tier — one subscription delivers productivity, identity, device management and endpoint security together.
Visual Studio Code	Latest stable release	Free (proprietary licence)	Primary code editor for the development team, with extensions for the project languages, linting and remote container development.
Git	Latest stable release	Free and open source (GPLv2)	Distributed version control underpinning all client project work, code review and release history.
GitHub Desktop	Latest stable release	Free and open source (MIT)	Graphical Git client for staff who are not comfortable at the command line, particularly QA and junior developers reviewing branches.
Oracle VirtualBox	Latest stable 7.x release	Base package free (GPLv3)	Local virtual machines for developers testing against clean Windows and Linux targets. See the licensing note below regarding the separately licensed Extension Pack.
Google Chrome	Latest stable release	Free (proprietary licence)	Standard browser for web application testing and access to cloud services, centrally configured using the Chrome Enterprise policy templates.
Microsoft Defender Antivirus	Built in, evergreen	Included with Windows 11 Pro	Endpoint anti-malware on every Windows device, centrally monitored and policy-managed through Microsoft Defender for Business.
AnyDesk	Latest stable release	Commercial licence required	Remote support tool allowing the IT Manager to assist staff without leaving the desk and to reach machines used off site. See the licensing note below.
7-Zip	Latest stable release	Free and open source (GNU LGPL)	Archive creation and extraction, including AES-256 encrypted archives used when files must be handed to a client outside the normal channels.
Bitwarden (Teams)	Latest stable release	Subscription, per user	Company password manager. It is what makes the password policy in Part 7.6 realistic, since staff cannot memorise a unique long passphrase for every service.
Veeam Backup & Replication Community Edition	Latest stable release	Free tier, up to 10 workloads	Backs up the server's virtual machines to the NAS and to the rotating external drives. The free tier's ten-workload limit comfortably covers the planned five virtual machines.

Table 4 — Enterprise software inventory and licensing.

3.1 Licensing Considerations

VirtualBox Extension Pack is not free for business use.

The VirtualBox base package is licensed under the GPL and may be used commercially at no cost. The Extension Pack is a separate product under Oracle's Personal Use and Evaluation Licence, which expressly excludes use in a business. ABC Startup Solutions will therefore deploy the base package only, and will purchase the Oracle Enterprise licence if a specific feature such as USB 2.0/3.0 passthrough is later shown to be necessary. This is one of the most commonly overlooked licence conditions in small IT environments.

AnyDesk's free version is licensed for personal use only.

AnyDesk's free offering is limited to private, non-commercial use, and the software detects and interrupts sessions it identifies as commercial. A paid licence is budgeted for the IT Manager rather than relying on the free version, which would be both a licence breach and an unreliable support tool.

Subscription counts must be tracked against headcount.

Microsoft 365 and Bitwarden are billed per user. The joiner–mover–leaver process in Part 7.4 includes reclaiming licences on departure, which prevents the company from quietly paying for accounts that no longer belong to anyone.

Microsoft 365 Business Premium is capped at 300 users.

This is well beyond the company's growth horizon, but it is recorded here so that a future administrator knows the ceiling exists and that crossing it requires a move to an enterprise plan.

Part 4 — Enterprise Network Inventory

Network equipment, structured cabling, port budget and VLAN plan

Asset ID	Network Equipment	Qty	Location	Purpose
NET-MDM-001	ISP Fibre Modem / ONU	1	Server cabinet	Terminates the fibre service from the provider. Operated in bridge mode so that routing, NAT and firewalling stay on equipment the company controls, which means the unit can be swapped or the provider changed without redesigning the network.
NET-RTR-001	Business Edge Router (dual WAN)	1	Server cabinet	Handles WAN termination and PPPoE authentication, NAT, static routing and the site-to-site and client VPN endpoints. The second WAN port carries the 5G/LTE backup service and fails over automatically.
NET-FW-001	UTM Firewall Appliance	1	Server cabinet	Security gateway providing stateful filtering, intrusion prevention, web and application control, and TLS inspection. It also terminates every VLAN interface, so all traffic between departments is routed and filtered here rather than being switched freely.
NET-SW-001	48-Port Managed Gigabit PoE+ Switch	1	Server cabinet	Single access switch for the floor. Provides 802.1Q VLAN tagging, PoE+ for the access points, port security and a tagged trunk uplink to the firewall.
NET-AP-001	Wireless Access Point, Wi-Fi 6 (802.11ax), PoE+	2	Ceiling: open work area, meeting room	Wireless access for laptops and mobile devices. Each broadcasts a corporate SSID mapped to the Sales VLAN and an isolated guest SSID mapped to the guest VLAN.
NET-PP-001	24-Port CAT6 Patch Panel	2	Server cabinet	Forty-eight termination positions matching the switch exactly. Every horizontal run is punched down to a panel instead of being terminated on a plug, so a faulty patch cord can be swapped without re-terminating solid-core cable.
NET-CBL-001	CAT6 U/UTP Solid Cable, 305 m box	3	Structured cabling, floor-wide	Approximately 26 horizontal runs are planned: 23 active drops for the desks, access points and printer identified in the port budget, plus 3 spare drops reserved for the meeting room and reception during the initial fit-out. At an average of roughly 25 m per run this represents about 650 m of horizontal cable; three boxes provide roughly 915 m, covering routing detours, service loops, terminations and waste. The spare drops give installation flexibility and avoid disruptive cable work during future expansion.
NET-CBL-002	CAT6 Stranded Patch Cords (1 m / 2 m)	40	Server cabinet and workstations	Factory-made cords for panel-to-switch patching and desk outlet-to-device connections, which are more reliable than hand-terminated cords at the point of most frequent handling.
NET-RJ45-001	CAT6 RJ45 Connectors (shielded, box)	200	IT stock	Field terminations for access point tails, replacement patch cords, re-terminations and test leads. Wall outlets and patch panels use punch-down keystone jacks rather than plugs, so far fewer connectors are consumed than a naive two-per-run estimate would suggest.

Table 5 — Enterprise network equipment inventory.

4.1 Switch Port Budget

The switch size was chosen by counting connections rather than by counting employees. A 24-port switch would have been fully consumed on the first day with three connections left unplugged.

Connected Device	Ports
Employee desk drops (one per workstation)	20
Application server (two bonded NICs)	2
NAS storage	1
Network printer / MFP	1
Wireless access points (PoE+)	2
Tagged trunk uplink to the UTM firewall	1
Total in use at go-live	27
Spare capacity for growth	21
Installed capacity	48

Table 6 — Switch port allocation at go-live.

4.2 VLAN Segmentation Plan

VLAN	Name	Subnet	Assigned To	Access Policy
10	IT	192.168.10.0/24	IT Department (5 users)	Full administrative access to VLAN 50; the only segment permitted to reach device management interfaces.
20	HR	192.168.20.0/24	Human Resources (4 users)	Internet, Microsoft 365 and printing. No access to Finance systems or to other user VLANs.
30	Finance	192.168.30.0/24	Finance (5 users)	The most restricted segment. Accepts no inbound sessions from any other user VLAN; outbound access limited to the internet, Microsoft 365 and printing.
40	Sales	192.168.40.0/24	Sales & BD (6 users), corporate Wi-Fi	Internet, Microsoft 365, printing and the shared sales folder on the NAS.
50	Servers & Infrastructure	192.168.50.0/24	Server, NAS, printer, management interfaces	Reachable only on specific service ports from the user VLANs; full access from VLAN 10.
60	Guest Wi-Fi	192.168.60.0/24	Visitor devices	Internet only, with client isolation enabled and no route to any internal VLAN.

Table 7 — VLAN plan and inter-segment access policy.

4.3 Network Design Logic

Traffic enters the building on the fibre service and terminates on the provider's modem, which is run in bridge mode. Bridging it is a deliberate choice: it keeps NAT, routing, firewall policy and VPN configuration on equipment the company owns and can document, so changing internet provider becomes a matter of re-authenticating one WAN interface rather than rebuilding the network.

The edge router performs WAN termination, NAT and VPN, and holds the second WAN interface for the 5G/LTE backup service. Keeping the router distinct from the firewall means the connectivity function and

the security function fail independently, and a router problem can be worked around temporarily without dismantling the security policy.

The UTM firewall sits behind the router and is the most important device in the design, because every VLAN interface terminates on it. The switch is a layer-2 device and does not route, so traffic from Finance to the server, or from HR to Sales, has no path except through the firewall, where it meets a written policy and is logged. This is what turns VLANs from a cosmetic label into an actual security control; VLANs alone separate broadcast domains but prove nothing about who can reach what.

The managed switch carries all 27 live connections, tags each access port into its department VLAN, and delivers PoE+ power to the two access points. Its uplink to the firewall is a single 802.1Q trunk carrying all six VLANs.

The access points broadcast two SSIDs. The corporate SSID uses WPA3-Enterprise authentication against Microsoft Entra ID and places clients into the Sales VLAN. The guest SSID is mapped to VLAN 60, has client isolation enabled, is bandwidth-limited and is permitted to reach the internet only — a visitor's laptop can never see a company device.

The design was checked for unnecessary complexity. Six VLANs is the smallest number that still separates the two confidential departments, the infrastructure and untrusted guests. There is no core/distribution/access hierarchy, no stacking, no redundant switch and no dynamic routing protocol, because at 20 users on one floor none of those would earn the configuration and troubleshooting burden they create for a single administrator.

Part 5 — Enterprise Network Diagram

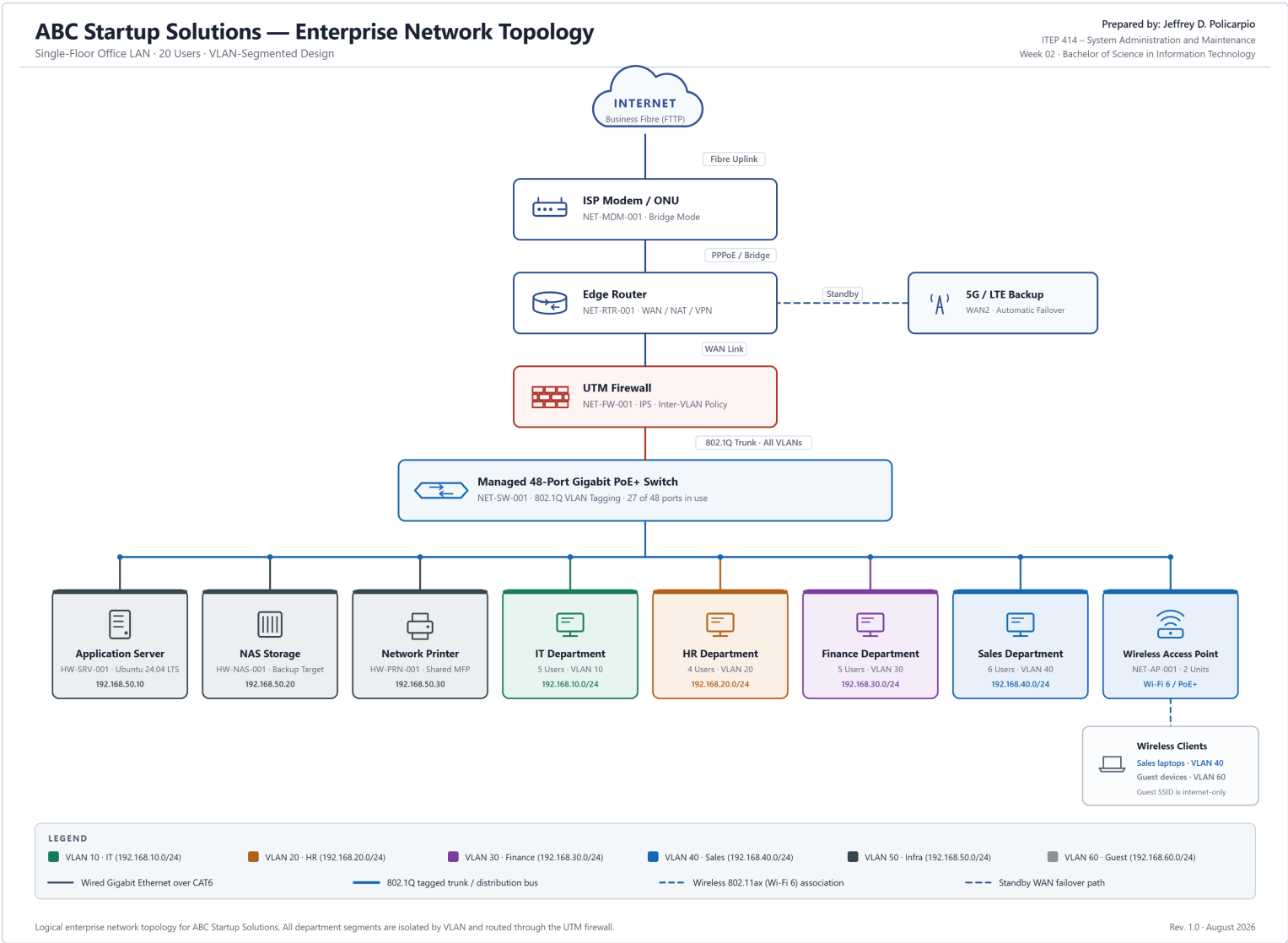


Figure 2 — Logical enterprise network topology for ABC Startup Solutions. Full-resolution PNG, PDF and editable Draw.io copies are provided in the Week02/diagrams folder of the repository.

5.1 Reading the Diagram

Figure 2 shows the logical topology. Reading it from the top: the internet arrives over business fibre, terminates on the bridged ISP modem, passes to the edge router for NAT and VPN, and then to the UTM firewall, which inspects north–south traffic and routes between the department VLANs. A single 802.1Q trunk links the firewall to the managed switch, which distributes all wired connections.

Below the switch, the eight branches are drawn in two groups. The three grey branches on the left are infrastructure on VLAN 50: the application server, the NAS and the shared network printer. The four coloured branches in the middle are the department segments, each on its own VLAN and subnet. On the right, the wireless access points connect over PoE+ and carry both the corporate Sales SSID and the isolated guest SSID.

The dashed link from the 5G/LTE backup unit to the router's second WAN port is a standby path. It carries no traffic while the fibre service is healthy and is brought up automatically when the primary circuit fails.

A practical example of how the diagram enforces policy: when a Finance workstation on VLAN 30 prints a payroll summary, the job does not travel directly to the printer even though both devices are plugged into the same physical switch. It is tagged into VLAN 30, carried up the trunk to the firewall, matched against a rule that permits only printing protocols from VLAN 30 to the printer's address on VLAN 50, and only then delivered. The same traffic attempting to reach a Sales workstation would be dropped and logged.

Part 6 — System Administration Roles

Responsibilities, skills, tools and certifications for four core roles

6.1 Helpdesk Technician

The Helpdesk Technician is the first point of contact between employees and the IT function. The role exists to resolve the high volume of routine problems quickly and to make sure the ones that cannot be resolved reach the right specialist with enough information attached to be useful.

Responsibilities

- Receive, log, categorise and prioritise support requests, and keep the requester informed until the ticket is closed.
- Resolve first-line issues: password and account lockouts, printing failures, email and connectivity problems, and application errors.
- Prepare, image and issue new workstations, and configure peripherals and mobile devices.
- Carry out user onboarding and offboarding, including account creation, group membership, licence assignment and the return and wiping of equipment.
- Maintain the hardware and software asset register and record licence allocations.
- Escalate to network, server or cloud specialists with clear reproduction steps and diagnostic evidence.
- Write and maintain knowledge-base articles so recurring problems become self-service.

Skills

- Windows and macOS desktop troubleshooting; a working knowledge of common office applications.
- Account administration in Active Directory or Microsoft Entra ID: resets, group membership, licence assignment.
- Basic networking: confirming an IP address, testing DNS, distinguishing a local fault from a site-wide one.
- A methodical diagnostic approach — reproduce, isolate, test one variable at a time.
- Clear written and verbal communication, patience, and the discipline to document while working rather than afterwards.
- Awareness of when a request is a security event, such as a suspected phishing message, rather than an ordinary fault.

Common Tools

- Service desk platforms: Jira Service Management, Freshservice, GLPI, osTicket.
- Remote support: AnyDesk, Windows Quick Assist, TeamViewer.
- Administration consoles: Microsoft 365 admin center, Microsoft Entra admin center, Microsoft Intune.
- Deployment and imaging: Windows Autopilot, Microsoft Deployment Toolkit.
- Knowledge base and documentation: SharePoint, Confluence, internal wikis.

Relevant Certifications

Certification	Focus
CompTIA A+	The standard entry credential for hardware, operating systems and troubleshooting.
CompTIA Network+	Networking fundamentals, useful for judging when to escalate.
Microsoft 365 Certified: Endpoint Administrator Associate (MD-102)	Managing and securing Windows endpoints with Intune.
ITIL 4 Foundation	Service management vocabulary and process — incident, request and problem handling.

6.2 Network Administrator

The Network Administrator is responsible for the connectivity every other system depends on. The role covers designing, configuring, monitoring and securing the wired and wireless network, and it owns the boundary between the organisation and the internet.

Responsibilities

- Design and document the LAN, WLAN and WAN, including addressing plans, VLAN assignments and topology diagrams.
- Configure and maintain switches, routers, firewalls and wireless controllers, and manage firmware lifecycle.
- Write and review firewall rules and access control lists, applying least privilege between segments.
- Operate remote access: site-to-site and client VPN, certificates and authentication integration.
- Manage core network services — DHCP scopes, internal DNS, NTP.
- Monitor availability, bandwidth utilisation and error rates, and plan capacity before saturation causes complaints.
- Troubleshoot faults from the physical layer upward, including cable certification and wireless interference surveys.
- Act as the technical contact for the internet provider and hold them to the service level agreement.

Skills

- TCP/IP, IPv4 subnetting and a working understanding of IPv6.
- Switching: VLANs, 802.1Q trunking, spanning tree, link aggregation, port security.
- Routing: static routes, and dynamic protocols such as OSPF where the scale justifies them.
- Firewall policy design, network address translation and VPN technologies.
- Wireless: channel planning, RF coverage, WPA2/WPA3 and 802.1X authentication.
- Packet-level analysis to settle arguments about whether a fault is the network's.
- Scripting and automation for repeatable configuration and backup of device configurations.

Common Tools

- Vendor command lines and consoles: Cisco IOS, FortiGate, MikroTik RouterOS, Ubiquiti UniFi.
- Analysis: Wireshark, tcpdump, iperf3, Nmap.
- Monitoring: Zabbix, LibreNMS, PRTG, SolarWinds; NetFlow collectors.

- Design and simulation: Cisco Packet Tracer, GNS3, EVE-NG, draw.io for topology documentation.
- Physical layer: cable certifiers, tone generators, wireless site survey software.

Relevant Certifications

Certification	Focus
Cisco CCNA	The widely recognised associate-level networking certification, covering switching, routing, security fundamentals and automation.
CompTIA Network+	Vendor-neutral networking foundation.
CompTIA Security+	Security fundamentals that firewall and access control work depends on.
Cisco CCNP Enterprise	Professional-level progression for larger or more complex networks.

6.3 Linux System Administrator

The Linux System Administrator builds and keeps running the Linux servers that host internal and customer-facing services. In a software company the role sits close to the development team, because the servers being maintained are the ones the developers build and deploy on.

Responsibilities

- Install, configure, patch and decommission Linux servers, physical and virtual.
- Manage users, groups, file permissions and privilege escalation through sudo policy.
- Administer storage: partitioning, LVM, filesystems, quotas and capacity growth.
- Run core services — web servers, databases, DNS, mail relays, application runtimes.
- Automate repetitive work with shell scripting and configuration management so servers are reproducible rather than hand-built.
- Configure and verify backups, and rehearse restores rather than assuming they work.
- Monitor performance and analyse logs to find the cause of faults, not just the symptom.
- Harden systems: SSH key-only authentication, host firewalls, SELinux or AppArmor, and prompt patching of security advisories.

Skills

- Fluency at the command line and in shell scripting (Bash).
- systemd service management, journald log analysis, and boot and recovery troubleshooting.
- Package management across families: APT on Debian and Ubuntu, DNF on RHEL derivatives.
- Networking from the host's side: interface configuration, routing tables, name resolution, firewall rules.
- Storage and filesystem management, including LVM and RAID concepts.
- Performance analysis — identifying whether a system is constrained by CPU, memory, disk or network.
- Virtualisation and containers: KVM, Docker, and increasingly Kubernetes.
- Version control for configuration, and a habit of documenting what was changed and why.

Common Tools

- Shell and core utilities: Bash, ssh, rsync, cron and systemd timers, vim.

- Configuration management and automation: Ansible, Git.
- Virtualisation and containers: KVM/libvirt, Docker, Podman.
- Monitoring and observability: Prometheus with Grafana, Zabbix, Nagios; journalctl and centralised logging.
- Diagnostics: top and htop, iostat, ss, dmesg, strace.
- Backup: Veeam Agent for Linux, restic, Bacula.

Relevant Certifications

Certification	Focus
Red Hat Certified System Administrator (RHCSA, EX200)	A fully hands-on, performance-based exam and the most recognised Linux administration credential in enterprise environments.
Linux Foundation Certified System Administrator (LFCS)	Vendor-neutral and also performance-based, solving real problems at a live command line.
LPI LPIC-1 (exams 101-500 and 102-500)	Distribution-neutral foundation covering system architecture, administration and essential commands.
CompTIA Linux+ (XK0-006)	Entry-level Linux administration in cloud and hybrid environments.

6.4 Cloud Administrator

The Cloud Administrator manages the organisation's resources on public cloud platforms. The role covers provisioning and operating those resources, controlling who may do what, keeping the monthly bill defensible, and making sure that workloads running outside the office are as well protected as the ones inside it.

Responsibilities

- Provision and operate cloud compute, storage, database and networking resources.
- Design and administer identity and access management: roles, policies, service principals and least-privilege permissions.
- Define and enforce cloud network topology — virtual networks, subnets, security groups and hybrid links back to the office.
- Monitor cost, set budgets and alerts, right-size resources and remove those nobody is using.
- Configure backup, replication and disaster recovery for cloud-hosted workloads.
- Apply security and compliance controls: encryption at rest and in transit, audit logging, policy guardrails.
- Deploy infrastructure as code so environments are reproducible and reviewable rather than clicked together in a console.
- Plan and execute migrations of on-premises workloads, and manage the hybrid period in between.

Skills

- Working knowledge of at least one major platform: AWS, Microsoft Azure or Google Cloud.
- Cloud identity and access management, and the discipline to grant the narrowest workable permission.
- Cloud networking concepts: virtual networks, routing, load balancing, private endpoints, VPN and peering.

- Infrastructure as code: Terraform, Bicep or CloudFormation.
- Scripting and automation in Python, PowerShell or Bash, and comfort with each platform's CLI.
- Cost management and the architectural judgement to tell genuine elasticity from expensive idle capacity.
- Containers and orchestration, since a large share of cloud workloads now arrive packaged that way.
- A solid grounding in Linux and networking — cloud administration assumes both rather than replacing them.

Common Tools

- Platform consoles and command lines: AWS Management Console and AWS CLI, Azure portal with Azure CLI and PowerShell, Google Cloud console with gcloud.
- Infrastructure as code: Terraform, AWS CloudFormation, Azure Bicep.
- Monitoring and logging: Amazon CloudWatch, Azure Monitor, Google Cloud Operations Suite.
- Identity: Microsoft Entra ID, AWS IAM, Google Cloud IAM.
- Containers: Docker, Kubernetes and the managed services EKS, AKS and GKE.

Relevant Certifications

Certification	Focus
AWS Certified Cloud Practitioner	Foundational credential covering cloud concepts, AWS services, security and pricing.
AWS Certified CloudOps Engineer – Associate (SOA-C03)	The operations-focused associate certification, renamed from AWS Certified SysOps Administrator – Associate; it validates deployment, monitoring, scaling and security operations on AWS.
Microsoft Certified: Azure Administrator Associate (AZ-104)	Implementing, managing and monitoring an Azure environment, including identity, storage, compute and virtual networking.
Google Cloud Associate Cloud Engineer	Deploying, monitoring and maintaining projects and solutions on Google Cloud.

6.5 How the Four Roles Work Together at ABC Startup Solutions

Inside a company the size of ABC Startup Solutions these four roles are not four separate people, but they are four distinct sets of work that still have to happen, and the handovers between them are easiest to see by following a single complaint. A Finance associate reports that the accounting system has become unusably slow. The helpdesk work comes first: confirm whether it is one person or the whole department, check whether the workstation is short of memory or mid-update, and try the obvious fixes. If only that one user is affected the ticket usually ends there. If all five Finance staff are affected, the problem has outgrown the desk and the network work begins — checking the switch port and uplink for errors and saturation, confirming that the firewall rule permitting VLAN 30 to reach the server is not being hit by an unexpected volume of traffic, and ruling out a failing cable or a saturated internet circuit. If the network is clean, the question moves to the server: the Linux side looks at whether the virtual machine is starved of CPU or memory, whether the disk is filling, what the logs recorded when the slowdown started, and whether last night's backup job is still running and competing for the same disks. Should the eventual finding be that the server has simply outgrown the single host, the conversation becomes a cloud one — whether to move that workload to a cloud virtual machine, what it would cost each month, how identity

and access would be handled, and how the office would reach it securely over the VPN. The reason this matters in practice is that the handovers are where tickets die. A helpdesk note that says only 'Finance is slow' forces whoever picks it up next to start from nothing, whereas one that records who is affected, when it started and what has already been ruled out lets the next person begin where the last one stopped. At ABC Startup Solutions the IT Manager will wear all four hats in the early years, which makes the discipline of documenting each stage a note to their own future self as much as to a colleague — and it is what makes hiring a second person possible later without handing them an undocumented environment.

Part 7 — Infrastructure Recommendations

Practical recommendations with the reasoning and the benefit for each

7.1 Internet Provider

Recommendation. Subscribe to a business-grade symmetrical fibre plan in the region of 200–300 Mbps from an established Philippine provider such as PLDT Enterprise or Converge ICT, specifying a business service level agreement and at least one static public IP address. Add a 5G/LTE business SIM on the router's second WAN port as automatic failover.

Reason. Twenty staff running video calls, cloud file synchronisation, package and container downloads and nightly off-site backup need consistent upstream bandwidth, which is why a symmetrical business plan matters more than a large download figure on a consumer plan. A static IP is required for the site VPN. Both providers publish SME fibre offerings; Converge markets its FlexiBIZ range and PLDT its Beyond Fibre business service. Specific plan tiers and prices change frequently and must be confirmed directly with the provider at the time of purchase — no provider has been contacted and nothing has been purchased for this academic plan.

Benefit. A business SLA converts an outage from an open-ended wait into a contractual response time, and the cellular failover means a fibre cut interrupts work for seconds rather than for a day. For a company whose staff cannot work without access to cloud services and client systems, this is the highest-value single line item in the plan.

7.2 Server Specifications

Recommendation. Deploy one rack-mountable server running Ubuntu Server 24.04 LTS as a KVM virtualisation host, specified as set out in Table 8 and hosting five virtual machines: source control and build agent, development environment, staging environment, internal services (monitoring, documentation and the service desk), and a backup coordinator.

Reason. Consolidating five workloads onto one properly specified host is far cheaper to buy, power and administer than five machines, and virtualisation means a workload can be snapshotted before a risky change and rolled back in minutes. Ubuntu Server LTS carries no licence cost, is supported until 2029, and uses the same skills the IT Manager needs for the rest of the estate. Windows Server was deliberately not chosen, because the functions that would justify it — identity, email, file sharing — are already provided by Microsoft 365.

Benefit. One host to patch, monitor and back up instead of five; roughly half the memory and 44 per cent of the switch ports left free for growth; and no Windows Server licensing or client access licences to buy, track and renew.

Component	Specification	Reason
Processor	Intel Xeon E-2400 series, 6 cores / 12 threads (or AMD EPYC 4004 equivalent)	Enough parallel capacity for five concurrent virtual machines plus build jobs, without paying for a dual-socket platform the company cannot load.
Memory	64 GB DDR5 ECC (2 × 32 GB), expandable to 128 GB	Memory, not CPU, is the first constraint on a virtualisation host. ECC is specified because it corrects the single-bit errors that otherwise cause silent data corruption. Half the slots are left free so capacity can be doubled without replacing modules.

Component	Specification	Reason
Boot storage	2 × 480 GB SSD in RAID 1	A mirrored pair means a single boot drive failure does not take the host down.
Data storage	4 × 4 TB enterprise SATA in RAID 10 (≈8 TB usable)	RAID 10 is chosen over RAID 5 for better write performance under virtual machine workloads and a faster, lower-risk rebuild after a drive replacement.
RAID controller	Hardware controller with flash-backed write cache	Protects writes that are in flight when power is lost, which software RAID on a consumer board does not.
Network	2 × 1 GbE, bonded (LACP or active-backup)	Provides aggregate throughput and survives the loss of one cable, port or switch interface. 10 GbE would be unnecessary expense for 20 users.
Power	Dual redundant power supplies, both fed from the rack UPS	Removes the power supply as a single point of failure on the one host the company has.
Operating system	Ubuntu Server 24.04 LTS with KVM/libvirt	No licence cost, security maintenance to 2029, and a hypervisor the IT Manager can administer with the same Linux skills used everywhere else.
Form factor	Rack-mountable, installed in a lockable 12U wall-mounted cabinet	Keeps the server, NAS, switch, patch panels, firewall and router physically secured in the utility room.

Table 8 — Recommended server specification.

7.3 Backup Strategy

Recommendation. Implement a 3-2-1 backup strategy: three copies of the data, on two different types of media, with at least one copy off site. Set a recovery point objective of 24 hours and a recovery time objective of 8 hours for core services, and test a real restore every quarter with the result written down.

Reason. Primary data lives on the server and the NAS. The local backup on the NAS covers the common failures — a deleted file, a corrupted virtual machine, a bad deployment — and restores quickly because it never leaves the building. The rotating external drives cover what the NAS cannot: fire, theft, or ransomware that encrypts everything the server can reach over the network, which an unplugged drive is immune to by definition. The cloud copy exists because Microsoft 365 data is the company's most business-critical information and Microsoft's own retention is not a backup — a mailbox deleted beyond the retention window is gone.

Benefit. A single accidental deletion is recovered in minutes; a failed server is rebuilt within the working day; and a ransomware incident or a fire in the building is survivable with at most a week of lost work rather than the end of the company. The quarterly restore test is what separates a backup that works from a backup job that merely reports success — it is the only part of this section that proves the rest of it.

Copy	Location	Media	Contents	Frequency	Retention
Production data	Server and NAS	Internal disk arrays	Virtual machines, department file shares, source repositories	Live	—
Local backup	NAS (RAID 5)	Disk array	Image-level backup of all virtual machines and file shares	Nightly incremental, weekly full	30 daily, 12 weekly

Copy	Location	Media	Contents	Frequency	Retention
Off-site copy A	External USB drive, rotated	External HDD (offline)	Weekly full backup of virtual machines and shares	Weekly, drives swapped every Monday	12 weeks
Off-site copy B	Cloud object storage and Microsoft 365 backup	Cloud	Microsoft 365 mailboxes and OneDrive/SharePoint, plus critical repositories	Daily	12 months

Table 9 — 3-2-1 backup implementation. Recovery point objective 24 hours; recovery time objective 8 hours for core services.

7.4 Security Recommendations

Recommendation. Apply layered controls rather than relying on any single product: UTM firewall with intrusion prevention at the perimeter; VLAN segmentation with least-privilege rules between departments; Microsoft Defender for Business on every endpoint, centrally monitored; enforced patching through Intune update rings on Windows and unattended-upgrades on Ubuntu; multi-factor authentication on every cloud account without exception; separate administrative accounts so no one performs daily work with elevated rights; BitLocker on all Windows devices and LUKS on the server; WPA3 wireless with an isolated guest SSID; the server cabinet kept physically locked; a documented joiner–mover–leaver process; short annual security awareness training including phishing; and the 3-2-1 backup strategy set out in Section 7.3, treated here as a security control and recovery safeguard rather than only an availability measure.

Reason. Every one of these addresses a specific and likely failure. Segmentation limits how far one compromised workstation can reach. Multi-factor authentication defeats the credential theft that phishing is designed to achieve, and is the single highest-value control available to a company this size. Separating administrative accounts means malware caught while reading email does not inherit administrative rights. Disk encryption protects data on a laptop stolen from a client site or a taxi. The leaver process matters because an account that is never disabled is a permanent unmonitored way in. Awareness training is included because staff, not firewalls, are what phishing actually targets. Backups appear in this list deliberately: they are the last line of defence when every preventive control has failed, and a ransomware or destructive incident is survived by restoring, not by blocking.

Benefit. No single control failure results in a total compromise, and the company can demonstrate reasonable and appropriate organisational and technical measures for the personal information it processes, which is what the Data Privacy Act of 2012 requires of it as a personal information controller.

7.5 Antivirus and Endpoint Protection

Recommendation. Standardise on Microsoft Defender Antivirus, already built into Windows 11 Pro, and manage it centrally with Microsoft Defender for Business, which is included in the Microsoft 365 Business Premium subscription. Enable cloud-delivered protection, tamper protection and attack surface reduction rules, and configure alerts to reach the IT Manager. Do not install a second real-time antivirus product alongside it.

Reason. Defender is already licensed, requires no additional agent, and is tested independently against the same malware sets as commercial products. Defender for Business adds the part that actually matters operationally: a central console showing which machines are unprotected, out of date or currently alerting, instead of leaving each endpoint to report to nobody. Running two real-time scanners together is explicitly

counter-productive, since they intercept the same file operations and cause conflicts and severe performance loss.

Benefit. Full endpoint protection at no additional licence cost, one console for the entire fleet, and one less vendor relationship, agent and renewal for a single administrator to manage.

7.6 Password Policy

Recommendation. Adopt a policy aligned with current NIST guidance: a minimum of 15 characters for passwords used on their own and at least 8 where multi-factor authentication is enforced; passphrases encouraged over complex substitutions; every password unique to one service; screening against known-breached password lists; multi-factor authentication required on all cloud and remote access; account lockout or progressive rate limiting after repeated failures; no shared accounts, with any break-glass credential sealed and its use logged; separate privileged accounts for administrative work; and Bitwarden issued to every employee so that unique long passphrases are practical. Passwords are not expired on a fixed schedule; they are changed immediately when there is evidence of compromise.

Reason. NIST Special Publication 800-63B states that verifiers shall not require periodic password changes, and shall force a change only where there is evidence of compromise. Forced 90-day rotation reliably produces predictable variations such as an incremented digit, which weakens security while generating helpdesk tickets. Length beats complexity because it defeats brute force more effectively than substituting symbols for letters. Breach screening matters because a long password already published in a credential dump is worthless regardless of its length. The policy only works with a password manager, which is why Bitwarden is a line item in the software inventory rather than a suggestion.

Benefit. Materially stronger authentication with fewer helpdesk password resets, and a policy staff can actually follow — which matters because a policy people work around, by writing passwords on notes or reusing one everywhere, is worse than a modest policy they observe.

7.7 Expansion Plan

Recommendation. Scale in defined stages rather than rebuilding: to roughly 35 staff on the existing core equipment; then a second server host and a third access point; then a second site or a move of production workloads to the cloud.

Reason. The infrastructure was specified with measured headroom. Twenty-one of the 48 switch ports are unused, half the server's memory slots are empty, the NAS has room for larger drives, and Microsoft 365 Business Premium supports up to 300 users. Identity, email and file collaboration are already cloud-hosted and require no work at all to grow. That means the next 15 employees need workstations and licences but no new infrastructure.

Benefit. Growth becomes predictable and budgetable instead of arriving as an emergency, and the company avoids both of the usual mistakes — over-buying capacity that depreciates unused, and under-buying so that the network must be rebuilt in its second year.

Stage 1 — up to about 35 employees

Add workstations, monitors and Microsoft 365 licences only. Existing switch, firewall, router, server and NAS all absorb the load; add a third access point if the office floor area grows with the headcount.

Stage 2 — about 35 to 60 employees

Add a second server host and move to shared storage so virtual machines can be migrated between hosts, giving genuine high availability. Add a second 48-port switch and uplink it. Increase the internet plan and consider a second provider for diverse-path redundancy rather than only cellular failover.

Stage 3 — second location or production hosting

Extend the network to a second floor or site over a site-to-site VPN, keeping the existing VLAN scheme. Move client-facing production workloads to AWS or Azure, where they can scale independently of the office — the point at which the Cloud Administrator responsibilities described in Part 6.4 become a real part of someone's job.

Throughout

Formalise the service desk and asset register once headcount passes roughly 30, when tracking requests informally stops working, and hire a second IT staff member so the company is not dependent on one person's availability and undocumented knowledge.

Part 8 — Personal Reflection

Before this project I thought of IT infrastructure mostly as a list of equipment. If someone had asked me what a 20-person company needs, I would have answered with hardware — twenty computers, a server, a router — and considered the question answered. Working through this plan changed that. The equipment was the easy part. The difficult part was justifying every decision against the actual company: its size, its budget, the data it handles, and the fact that it can afford only one person to run everything.

The most valuable thing I learned is that good infrastructure decisions are mostly decisions about what not to build. My first draft included more than the company needed, because adding equipment felt like it made the design look more professional. When I went back and asked what each item actually solved, several came out. I had included a second server for redundancy before realising that the company's critical systems — email, files, identity — would already be in Microsoft 365, so the second server would have doubled the cost to protect the least critical workloads.

The most challenging task was the network design, specifically deciding whether to use VLANs. Segmentation is more work to configure and troubleshoot, and for one floor of twenty people a flat network would have worked. What convinced me was the Finance and HR data — payroll records and employment contracts sitting on the same flat network as everyone else. I also had to correct myself partway through: I thought putting departments on separate VLANs was itself the security control, and only later understood that VLANs separate broadcast domains, and that the real protection comes from forcing the traffic between them through the firewall. Getting the diagram to show that clearly took several attempts.

Planning before deployment stopped being an abstract idea. Almost every decision constrains a later one. A 24-port switch looked reasonable until I counted the ports properly and found 27 connections on day one. Cabling is even less forgiving, because running it is disruptive and expensive and cannot be corrected with a configuration change. Deciding these things on paper costs an afternoon; discovering them after installation costs money.

As a student aiming for a career in system administration, the habit I most want to carry forward is writing down the reason for a decision, not just the decision. Writing the collaboration section made me realise how much of this work is handing information to someone else — a colleague, or my own future self trying to remember why a firewall rule exists. I have no professional experience yet and this plan has not been deployed, but it is the first time I have had to defend a complete technical design from end to end.

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